DEEP IMPRESSIONS:

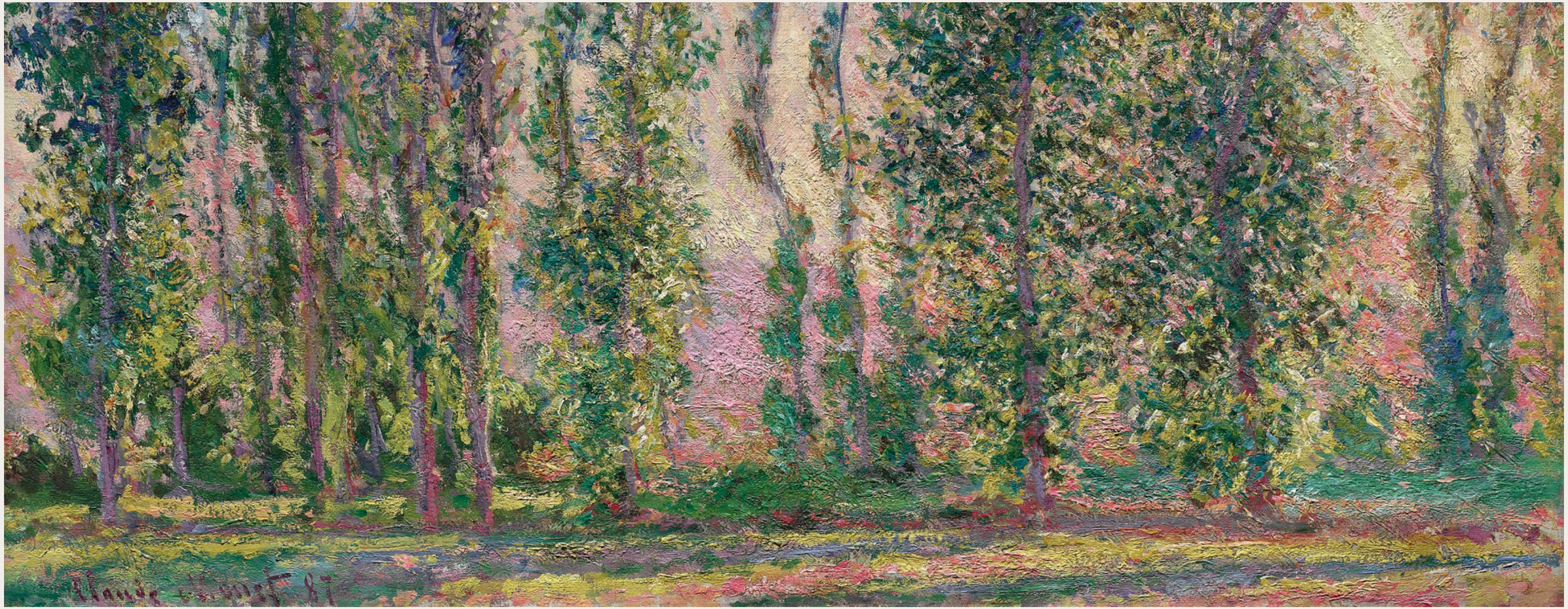
Training a Neural Network to Recognize Painters

Project 2 | Machine Learning: From Data to Predictions

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Machine Learning Bootcamp

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Poplars at Giverny (1887)
Monet

THE PROBLEM

Given an image of a painting, which of six painters made it?

Challenge

Stylistic Closeness

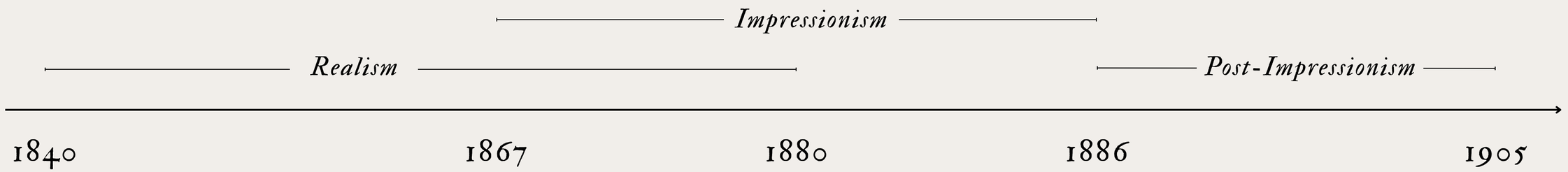
- **Realism → Impressionism:** Manet
- **Impressionism:** Monet, Renoir, Morisot
- **Post-Impressionism:** Cézanne, Van Gogh

Who is it for?

- **Catalogers and archivists** with large digitized collections, unlabeled or doubtful entries
- **Museum visitors and art students** training their eye for style

The model

- Confirms or correct a suggested painter
- Flags a label that disagrees with a confident prediction
- Explores who painted in a similar way



MANET

Bridge from Realism

- Never exhibited with the group, but inspired the younger painters
- More structured: flatter planes, strong contrasts, dark tones, clear outlines
- Where Impressionism blurs into Realism, a hard case for the classifier









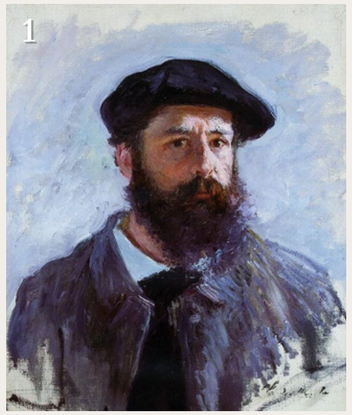


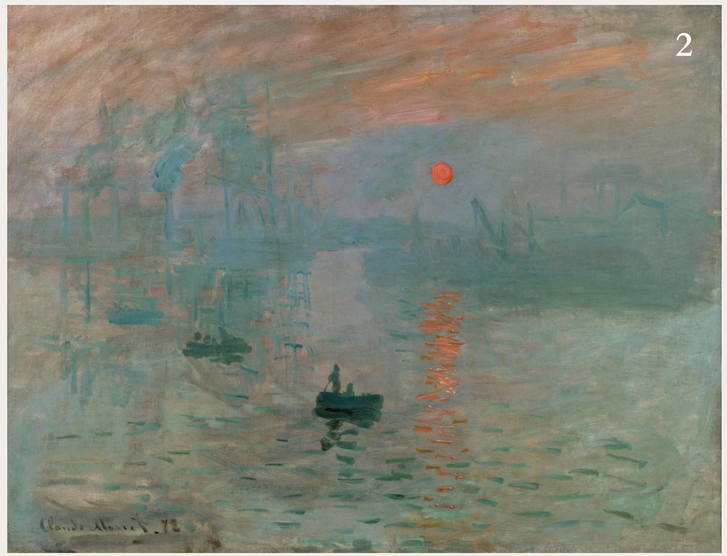
1 *Self-Portrait* (1879) · 2 *Boating* (1874) · 3 *The Fifer* (1866)
4 *Railway* (1873) · 5 *A Bar at the Folies-Bergère* (1882)

MONET

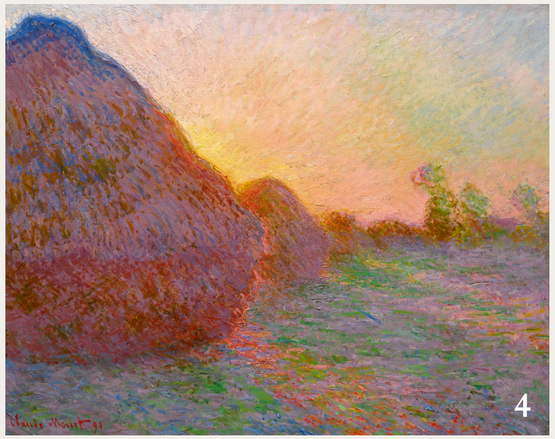
Light dissolving form

- Central figure; *Impression, Sunrise* named the movement
- Painted outdoors, chasing changing light (haystacks, Rouen Cathedral, water lilies)
- Broken, visible strokes, soft edges, color that dissolves form











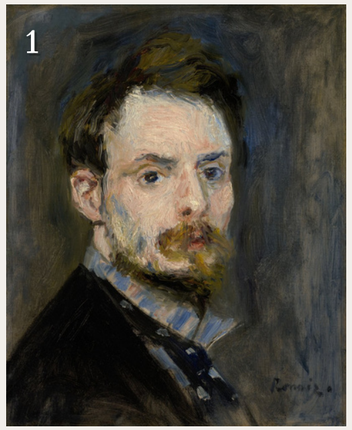


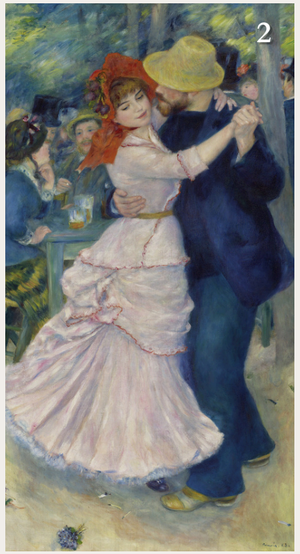
1 *Self-Portrait with a Beret* (1886) · 2 *Impression, Sunrise* (1872) · 3 *The Promenade, Woman with a Parasol* (1875)
4 *Haystacks (1890)* · 5 *Rouen Cathedral* (1892-93) · 6 *Water Lilies, Reflection of a Weeping Willows* (1916-19)

RENOIR

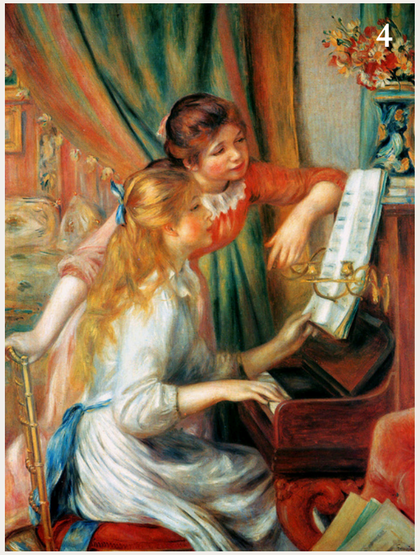
Warm, dappled sociability

- Shared Monet's light and loose brushwork; they painted side by side
- Focus on people: portraits, dances, bathers in dappled sunlight
- Warm, glowing color and soft, feathery strokes













1 *Self-Portrait* (1875) · 2 *Dance at Bougival* (1883) · *3 Umbrellas* (1885-86) · 4 *Girls at the Piano* (1892)
5 *Dance at Moulin de la Galette* (1876) · 6 *The Luncheon of the Boating Party* (1880-81)

MORISOT

Intimate, airy, sketch-like

- Founding member; exhibited in nearly all the group's shows
- Light, quick, sketch-like handling and a pale, airy palette
- Private subjects: domestic scenes, family, women's lives









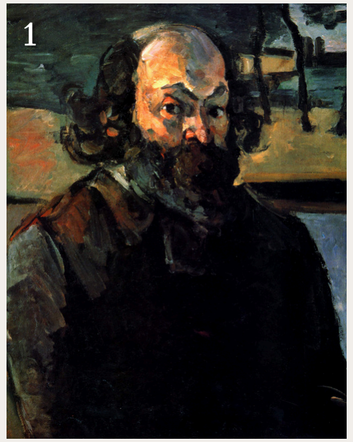


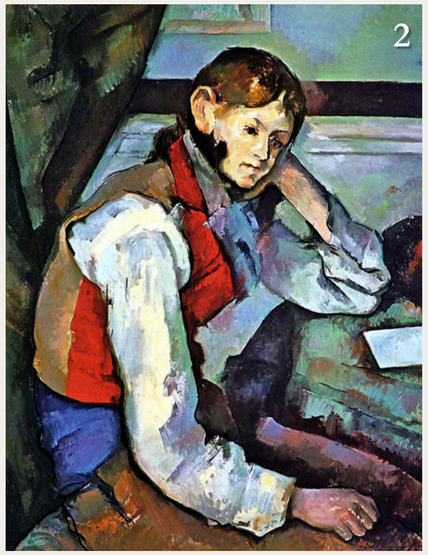
1 *Self-Portrait* (1885) · 2 *The Cradle* (1872) · 3 *Eugène Manet with His Daughter at Bougival* (1881)
4 *Summer's Day* (1879) · 5 *Young Woman at the Mirror* (1880)

CÉZANNE

Structure beneath color

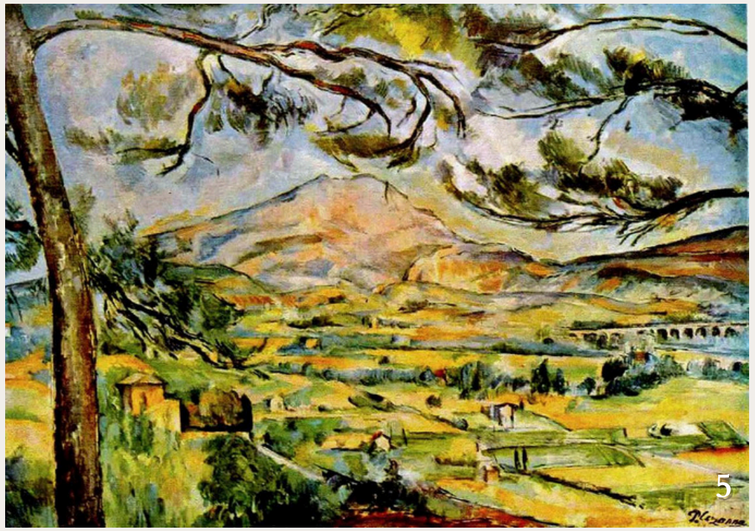
- Early exhibitor, but usually classed as Post-Impressionist
- Kept the color and landscapes, wanted solidity and order
- Constructive strokes and geometric forms, which later influenced Cubism









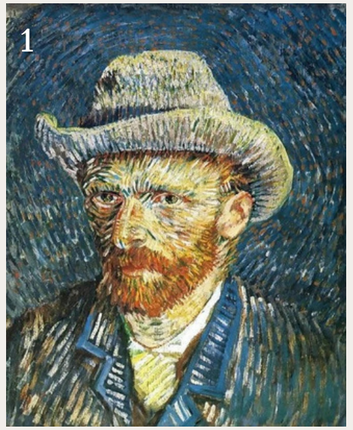


1 *Self-Portrait* (1875) · 2 *Boy in a Red Vest* (1889) · 3 *Basket of Apples* (1895)
4 *The Card Players* (1896) · 5 *Mont Sainte-Victoire* (1887)

VAN GOGH

Emotion in thick strokes

- Post-Impressionist: bright color and visible strokes, taken toward expression
- Thick impasto, swirling strokes, intense non-naturalistic color
- The most recognizable painter in the set, and the most distinct from Monet











1 *Self-Portrait with Felt Hat* (c. 1887) · 2 *Still Life - Vase with Fifteen Sunflowers* (1888) · 3 *The Starry Night* (1889)
4 *The Potato Eaters* (1885) · 5 *Bedroom in Arles* (1888)

THE DATA

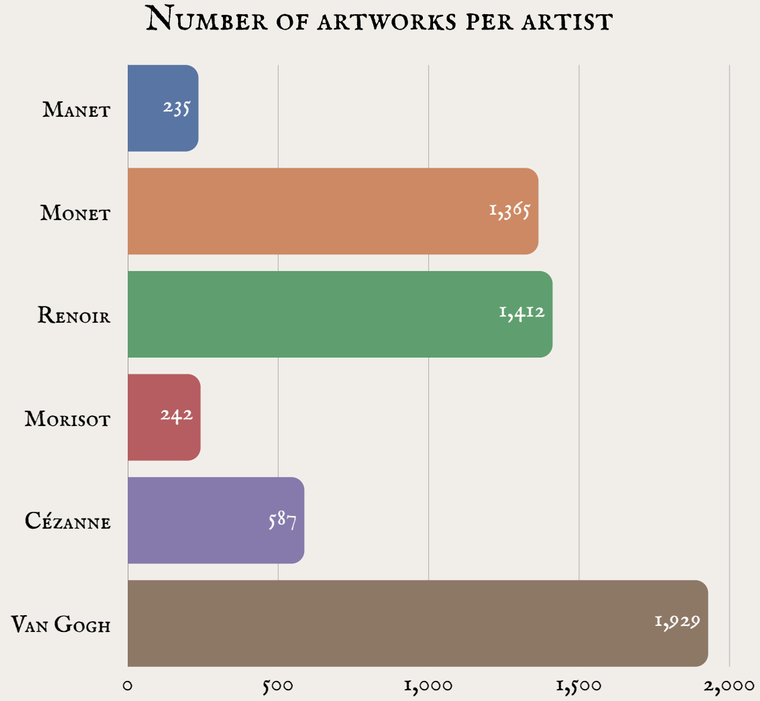
5,770 paintings, and the biggest trap was the same scene appearing twice

Dataset

- **Source:** WikiArt images
- **License**: public-domain paintings only
- **One row = one painting image**
  - File path, painter, title
- **Target = painter** (6 classes)
- **Class Imbalance of 8 to 1:**
  - Van Gogh 1,929 images vs. Manet 235
- **Split**: about 70/15/15, stratified by painter

Trap

- **The same work shows up many times**
  - Series, sketches, rescans
- A random split lets the model recognize a painting instead of a style.
- **Fix: perceptual hashing** (87 duplicate pairs found)



METRIC AND BASELINE

Always answering "Van Gogh" is right 34% of the time, so accuracy alone would flatter a model that learned nothing

Baseline

- **Model:** DummyClassifier predicting the most frequent painter
  - 5-fold cross validation
- **Score to beat:**
  - accuracy: **0.335**
  - macro-F1: **0.084**

Metric

Macro-F1

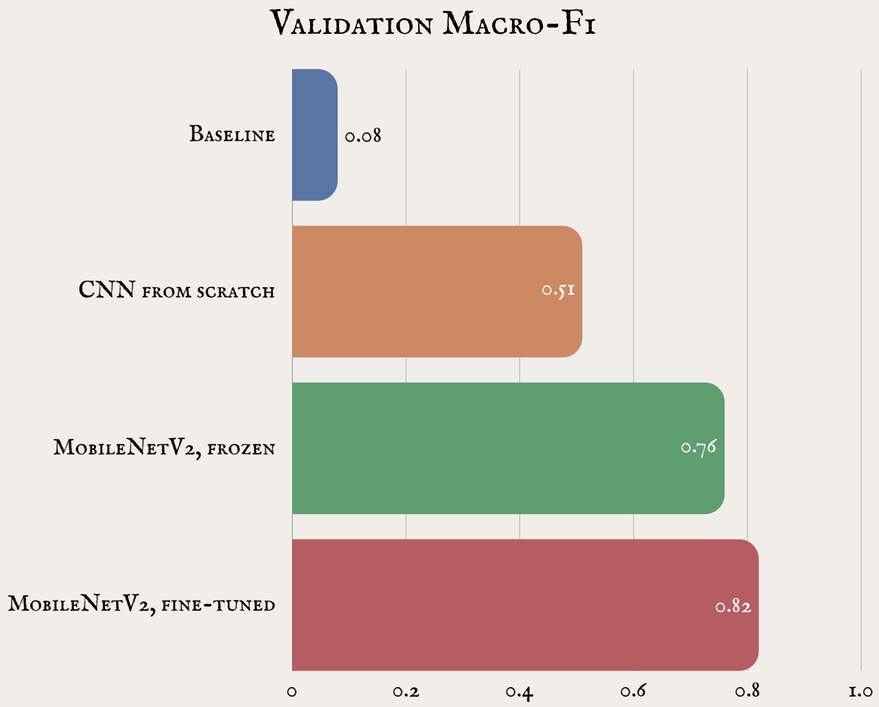
- It averages the F1 score of the six painters, so Manet and Morisot count as much as Van Gogh

Which mistake costs more?

- A confident wrong attribution misleads the reader more than an uncertain one, so I also track:
  - per-painter recall and the confusion matrix
  - the model's confidence on its errors

WHAT I TRIED

Pretrained features beat a from-scratch network by 0.25 macro-F1, and fine-tuning added 0.06 more



Models

- **Same setup for all models:**
  - Same splits, augmentation, and class-weighted loss.
  - **Early stopping** on validation loss, restoring the best weights.

1. Small **CNN** trained **from scratch**

2. **MobileNetV2, frozen**
   - Pretrained on ImageNet (1.2 million photos, 1,000 everyday categories), so it already sees edges, textures, and patterns.
   - Only a new last layer is trained on the paintings.

3. **MobileNetV2, fine-tuned**
   - The last 4 blocks are also trained at a very low learning rate

What did not not work

- Scratch CNN could not learn style properly
- The frozen model plateaued after about 9 epochs
- Fine-tuning overfits: training macro-F1 reached 0.94 against about 0.81 on validation, so I stopped on validation loss

THE FINAL MODEL

On 812 paintings it had never seen, the model names the right painter about 7 times in 8

Metric

- **Test set (812 paintings, evaluated once)**:
  - accuracy **0.873**, which is 709 correct and 103 wrong
  - macro-F1 **0.806** (95% CI 0.766 to 0.840)

- **Validation estimate: 0.820 macro-F1**, inside the test interval, so no sign of leakage or of overfitting to the validation set

Macro-F1

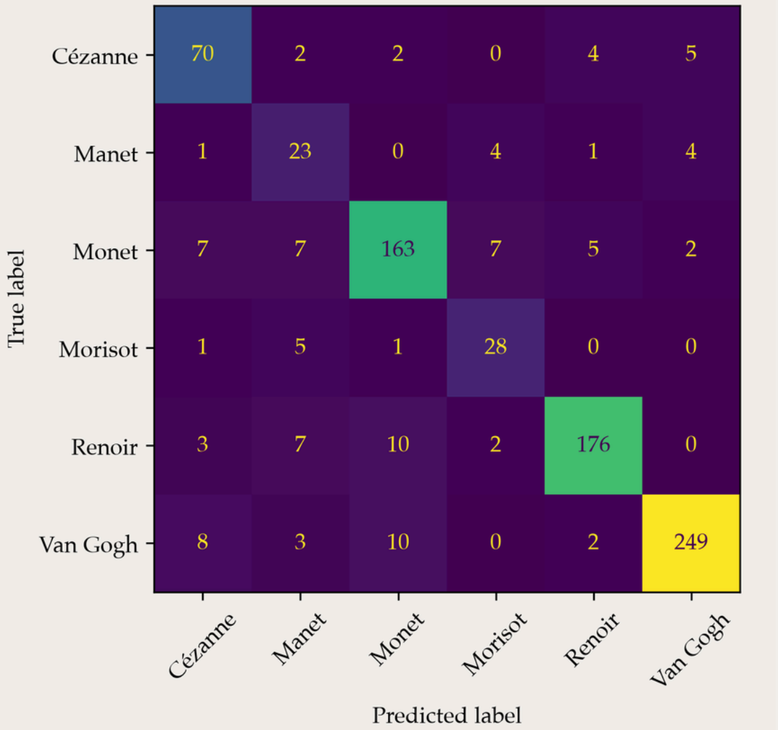
MobileNetV2 validation	MobileNetV2 test
0.820	0.806

Where it goes wrong

- **Mistakes follow shared technique:**
  - 10 Renoirs called Monet, the largest single group (light and loose brushwork, painted side by side)
  - Manet and Morisot swapped (4 and 5), both with loose figures
  - Van Gogh and Cézanne swapped (8 and 5), both with strong color
  - Monet's 28 errors are spread across every other painter (the reference point of the style)

- **Manet and Morisot have the worst results:**
  - Manet is predicted 47 times for 33 real paintings (precision 0.49).
  - Morisot is milder (precision 0.68).

Confusion Matrix



LIVE DEMO

Let's jump to
the Streamlit app!

CONCLUSIONS

Good for a first-pass suggestion, not for authentication

Where it succeded

- **Accuracy 0.87, macro-F1 0.81** on the test set
- **The estimate held up:** validation 0.82 sits inside the test interval
- **Transfer learning paid off**: 0.51 from scratch → 0.82 fine-tuned, gains for all six painters
- **Five of six painters at F1 of 0.74 or higher**
- **Remaining errors are plausible:** between painters with shared techniques

Where not to trust it

- **Manet:** about half of its “Manet” labels are wrong, so always send them to a person
- **Any painter outside the six:** it must pick one, so a Degas becomes a confident Monet or Renoir
- Photos of framed paintings, sketches, heavy crops: it only saw WikiArt reproductions
- **Its own confidence:** wrong answers still average about 0.7
- **Authentication or valuation:** 87% is not enough, and it sees broad style, not evidence

What would make it better

- **Make the confidence trustworthy:** choose the cutoff on validation data, and show the top-2 guesses with a reliability diagram
- **Fix the weak spots (Manet, Morisot)**:
  - Tune class weights
  - Add stronger augmentation
- **Preserve brushwork:** try higher-resolution inputs
- **Try other backbones**
- **Tune the fine-tuning:** unfreeze 2 blocks instead of 4, with more regularization



Wheatfield with Crows (1890)
Van Gogh

Thank you for your attention!